Chapter 7

THE APPLICATION LAYER

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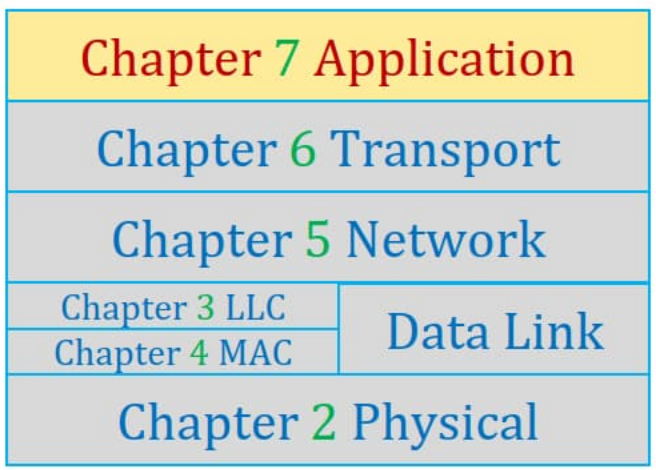

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7.1 DNS-The Domain Name System

- DNS: Domain Name System 域名系统
  - handles naming within the Internet
  - high-level, readable names
  - decouple machine names from machine addresses
- hosts.txt
  - a simply file
- DNS
  - a hierarchical, domain-based naming scheme and
  - a distributed database system for implementing this naming scheme.

7.1.1 The DNS Name Space

- ICANN: Internet Corporation for Assigned Names and Numbers
  - https://www.icann.org/ , manage the top of the naming hierarchy
- Top-level domains
  - two flavors: generic and countries
- Naming Rules
  - Each domain is named by the path upward from it to the root
  - The components are separated by periods (pronounced "dot")
  - Domain names are case-insensitive
  - Component names can be up to 63 characters long,
    - and full path names must not exceed 255 characters.
  - Domain names can be either absolute or relative
    - An absolute domain name always ends with a period.

Domain name space

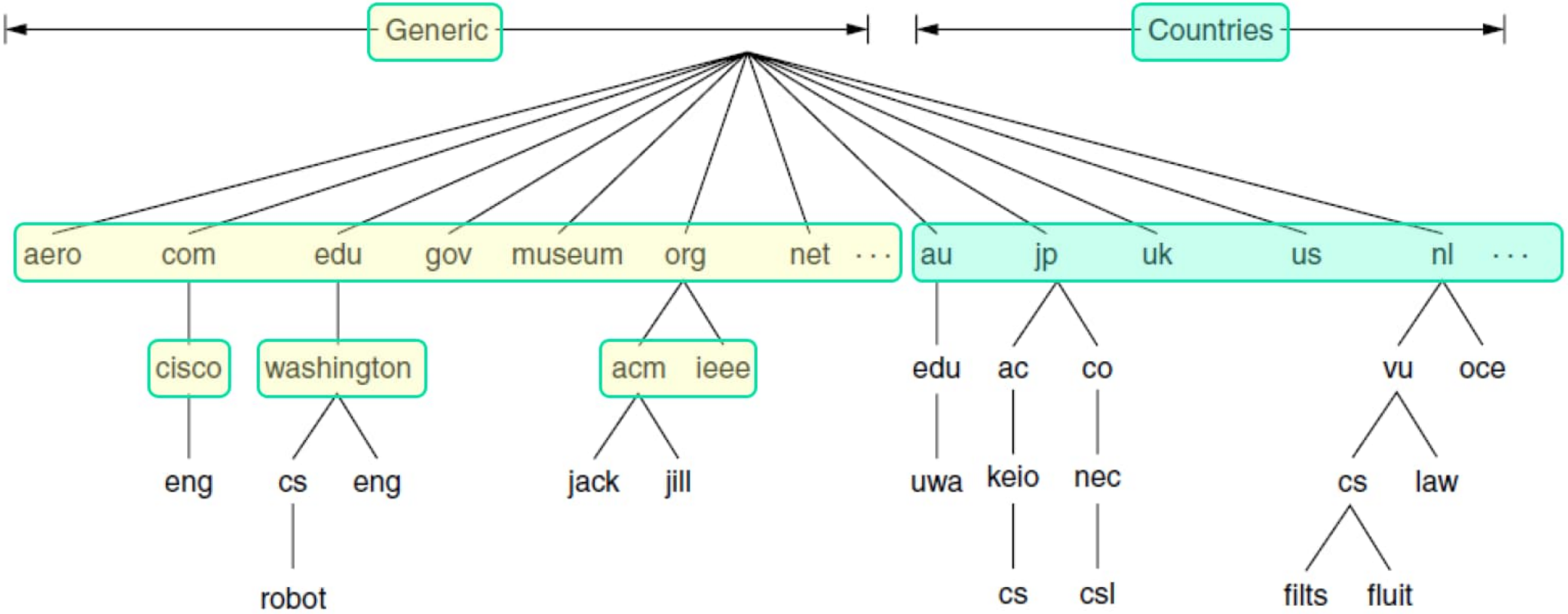

Figure 7-1. A portion of the Internet domain name space.

Generic top-level domains

Domain	Intended use	Start date	Restricted?
com	Commercial	1985	No
edu	Educational institutions	1985	Yes
gov	Government	1985	Yes
mil	Military	1985	Yes
net	Network providers	1985	No
org	Non-profit organizations	1985	No
int	International organizations	1988	Yes
aero	Air transport	2001	Yes
biz	Businesses	2001	No
coop	Cooperatives	2001	Yes
info	Informational	2002	No
museum	Museums	2002	Yes
name	People	2002	No
pro	Professionals	2002	Yes
cat	Catalan	2005	Yes
jobs	Employment	2005	Yes
mobi	Mobile devices	2005	Yes
tel	Contact details	2005	Yes
travel	Travel industry	2005	Yes
xxx	Sex industry	2010	No

7.1.2 Domain Resource Records

- Domain Resource Records 域资源记录
  - five-tuple (五元组): Domain_name, Time_to_live, Class, Type, Value

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Figure 7-3. The principal DNS resource record types.

DNS database

```
; Authoritative data for cs.vu.nl
; Domain_name, Time_to_live, Class, Type, Value
cs.vu.nl.      86400   IN   SOA      star boss (9527,7200,7200,241920,86400)
cs.vu.nl.      86400   IN   MX       1 zephyr
cs.vu.nl.      86400   IN   MX       2 top
cs.vu.nl.      86400   IN   NS       star

star           86400   IN   A        130.37.56.205
zephyr         86400   IN   A        130.37.20.10
top            86400   IN   A        130.37.20.11
www            86400   IN   CNAME    star.cs.vu.nl
ftp            86400   IN   CNAME    zephyr.cs.vu.nl

flits          86400   IN   A        130.37.16.112
flits          86400   IN   A        192.31.231.165
flits          86400   IN   MX       1 flits
flits          86400   IN   MX       2 zephyr
flits          86400   IN   MX       3 top

rowboat                IN   A        130.37.56.201
                       IN   MX       1 rowboat
                       IN   MX       2 zephyr

little-sister          IN   A        130.37.62.23

laserjet               IN   A        192.31.231.216
```

Figure 7-4. A portion of a possible DNS database for *cs.vu.nl*.

7.1.3 Name Servers

- zone 区域
  - the DNS name space is divided into nonoverlapping zones.

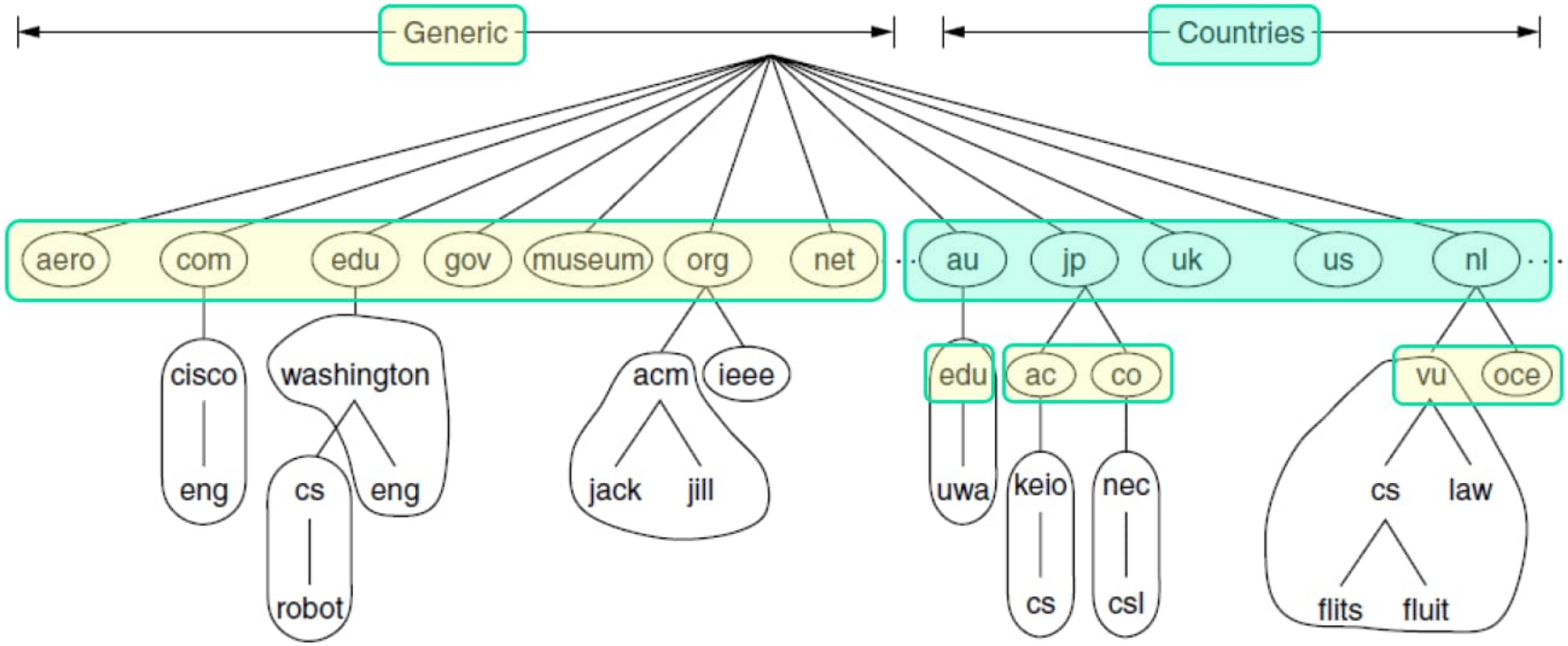

Figure 7-5. Part of the DNS name space divided into zones (which are circled).

Iterative query

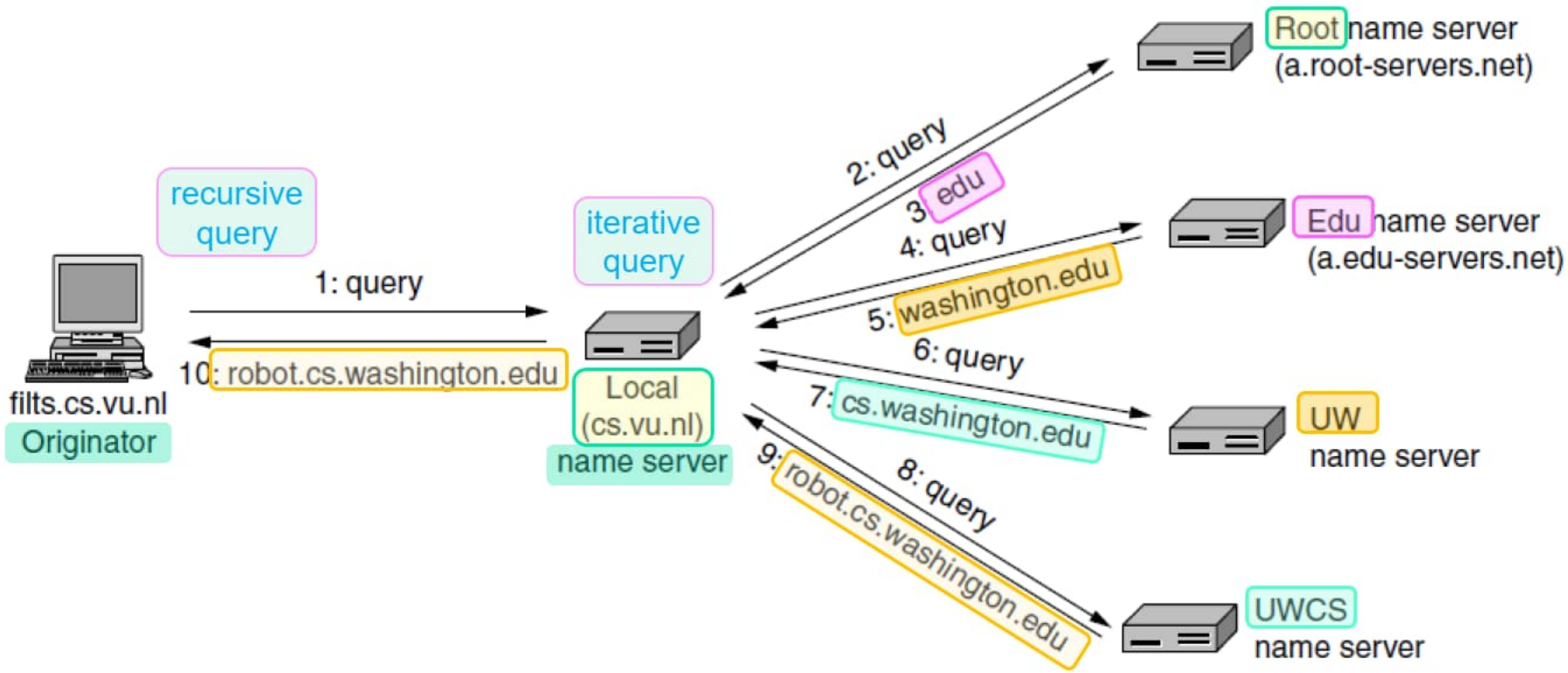

Figure 7-6. Example of a resolver looking up a remote name in 10 steps.

Summary

- Naming in the ARPANET started out in a very simple way:
  - an ASCII text file listed the names of all the hosts and their corresponding IP addresses.
- Domain Name System
  - a hierarchical scheme
  - At the top level are the well-known generic domains, including com and edu, as well as about 200 country domains.
  - implemented as a distributed database
  - query: recursive, iterative

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